

NEPTUNE

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NEPTUNE IS THE SMALLEST and the coldest of the four gas giants, as well as the most distant from the Sun. It was discovered in 1846, and just one spacecraft, Voyager 2, has been to investigate this remote world. When the

probe flew by in 1989, it provided the first close-up view of Neptune and revealed that it is the windiest planet in the solar system. Voyager 2 also found a set of rings encircling Neptune, as well as six new moons.

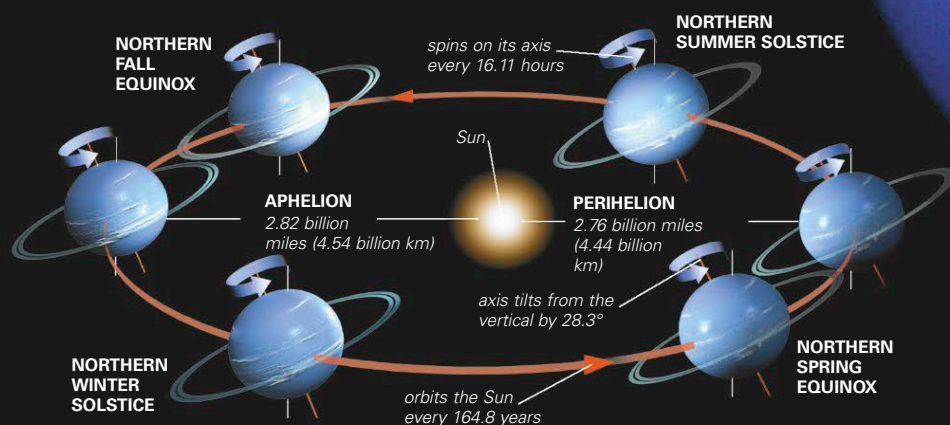
ORBIT

Neptune takes 164.8 Earth years to orbit the Sun, which means that it has completed only one circuit since its discovery in 1846. The planet is tilted to its orbital plane by 28.3° , and as it progresses on its orbit, the north and south poles point sunward in turn. Neptune is about 30 times farther from the Sun than Earth, and at this distance the Sun is 900 times dimmer. Yet this remote, cold world is still affected by the Sun's heat and light and apparently undergoes seasonal change. Ground-based and Hubble Space Telescope observations show that the southern hemisphere has grown brighter since 1980, and this—as well as an observed increase in the amount, width, and brightness of banded cloud features—has been taken as an indication of seasonal change. However, a longer period of observations is needed to be sure that this seasonal model is correct. The change is slow and the seasons

are long. The southern hemisphere is currently in the middle of summer. Once this is over, it is expected to move through fall, into a colder winter. Then, after 40 years of spring and a gradual increase in temperature and brightness, it will experience summer once more.

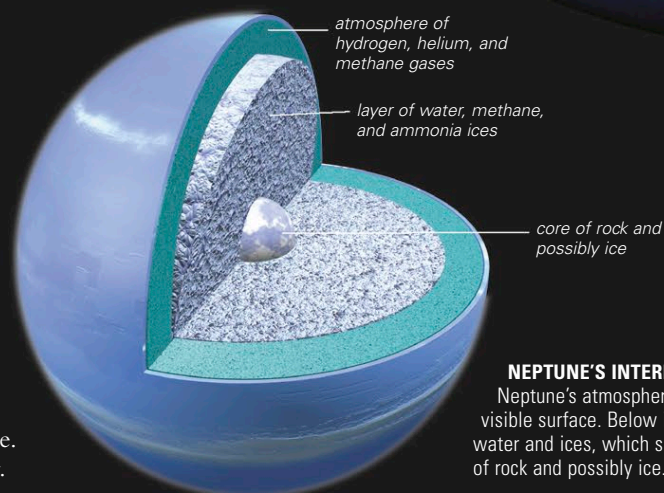
SPIN AND ORBIT

Neptune's orbit is elliptical, but less so than most planets. Only Venus has a more circular orbit. This means there is no marked difference between Neptune's aphelion and perihelion distances.



STRUCTURE

Neptune is very similar in size and structure to Uranus, and neither planet has a discernible solid surface. Like its inner neighbor, Neptune is too massive in relation to its size to be composed mainly of hydrogen. Only about 15 percent of the planet's mass is hydrogen. Its main ingredient is a mix of water, ammonia, and methane ices that makes up the planet's biggest layer. Neptune's magnetic field, which is tilted by 46.8° to the spin axis, originates in this layer. Above it lies the atmosphere. This is a shallow, hydrogen-rich layer that also contains helium and methane gas. Below the layer of water and ices, there is a small core of rock and possibly ice. The boundaries between the layers are not clearly defined. The planet rotates quickly on its axis, taking 16.11 hours for one spin, and as a result Neptune has an equatorial bulge. Its polar diameter is 527 miles (848 km) less than its equatorial diameter.



NEPTUNE'S INTERIOR

Neptune's atmosphere is the planet's visible surface. Below it lies a layer of water and ices, which surrounds a core of rock and possibly ice.